

Thinking inside the box: a greedy ball packing in the hypercube whose central radius appears to stabilize

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Abstract

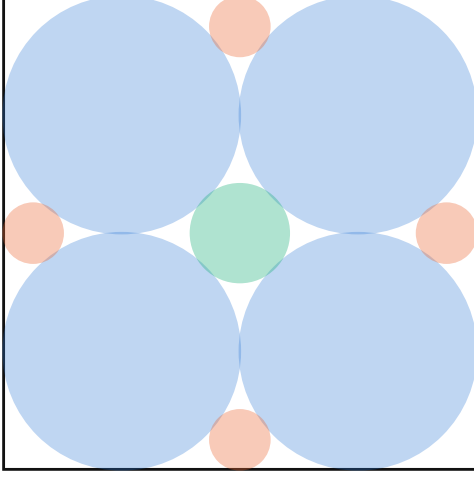
In the cube $[-1, 1]^N$, place a ball of radius $1/2$ at each corner. The largest ball centered at the origin that avoids them has radius $(\sqrt{N} - 1)/2$, which is unbounded and leaves the cube once $N > 9$. We study what happens when every face of the cube, of every dimension, receives a ball. Faces are processed in order of increasing dimension, and each ball is grown from radius 0 until its first contact with the cube, with an earlier ball, or with a ball of its own dimension. We prove that the radii satisfy an explicit recursion, with a same-dimension cap of $\sqrt{2} - 1$, and that the central radius is less than $1/2$ in every dimension, a value the corner-only radius already reaches at $N = 4$. Numerically, for $6 \leq N \leq 6000$ it takes only the two values $3 - 2\sqrt{2}$ and $\sqrt{2} - 1$, apart from $N = 9$ and $N = 17$ where it equals $1/3$. The two values alternate in mode-locked periods, the last occurrence of $\sqrt{2} - 1$ is at $N = 597$, and for every $598 \leq N \leq 6000$ the central radius is $3 - 2\sqrt{2}$. We explain the mechanism: in the regime observed numerically, the recursion reduces to a one-variable map with a repelling fixed point at $1/3$ (multiplier -2), a firing threshold just below it, and an exact reset to the cap. Trajectories for different N merge exactly when they reset on the same layer. We conjecture that the central radius equals $3 - 2\sqrt{2}$ for all $N \geq 598$, and we describe the obstruction to a simple proof.

1 Introduction

A well-known illustration of high-dimensional geometry, presented by Hamming [1, Ch. 9], places 2^N balls of radius 1 at the corners of $[-2, 2]^N$ and asks for the largest ball centered at the origin that avoids them. We work throughout in the cube scaled by $1/2$: balls of radius $1/2$ at the corners of $[-1, 1]^N$. The central radius is then $(\sqrt{N} - 1)/2$: it grows without bound, touches the facets of the cube at $N = 9$, and protrudes from the cube for $N > 9$. The “corners stick out” phenomenon behind it is a staple of introductions to high-dimensional geometry.

The corner balls leave most of the cube’s boundary structure unused. In this note every face of the cube receives a ball. A k -face is obtained by fixing $N - k$ coordinates to ± 1 ; there are $\binom{N}{k} 2^{N-k}$ of them, for $k = 0, 1, \dots, N$, the single N -face being the cube itself. Each face carries one N -dimensional ball, centered over the face and seated against the facets that contain it. Balls are placed greedily, one dimension at a time: corners first, then edges, and so on, ending with one ball at the center (Figure 1).

(a) $N=2$: $r_0 = \frac{1}{2}$, $r_1 = \frac{1}{8}$, $r_2 = \frac{\sqrt{2}-1}{2}$



(b) $N=3$: corners (faint), edges, faces, interior

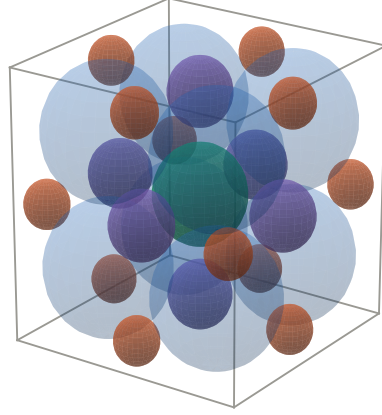


Figure 1: The construction for $N = 2$ and $N = 3$. For $N = 3$ the radii are $r_0 = \frac{1}{2}$, $r_1 = \frac{3-\sqrt{7}}{2} \approx 0.1771$, $r_2 = \frac{1}{4}$ and $r_3 = \frac{\sqrt{3}-1}{2}$; filling the faces of the cube in dimension 3 does not yet change the central ball.

Results.

1. *An exact recursion* (Theorem 6). Every ball on a k -face has the same radius r_k , with $r_0 = 1/2$; for $1 \leq k \leq N-1$, r_k is the minimum of the same-dimension cap $\sqrt{2}-1$ and explicit bounds from every lower layer; and the central radius r_N is a minimum over all lower layers.
2. *A uniform bound* (Proposition 8). For every $N \geq 2$, $r_N < 1/2$. The proof uses only the two layers directly below the center.
3. *A step map with a repelling fixed point* (Proposition 10). The bound that layer $k-1$ imposes on layer k is a map $f_p(r_{k-1})$, where $p = N-k$. For every $p \geq 2$, f_p fixes $1/3$ with derivative -2 , decreases every radius above $1/3$, and stops constraining the next layer altogether once its input falls below a threshold $x^*(p)$ with $1/3 - x^*(p) \sim 1/(9p)$.
4. *Numerical structure* (Section 6). For $16 \leq N \leq 6000$ every layer is either capped at $\sqrt{2}-1$ or pinned by the layer directly below it, so the recursion reduces to iterating $x \mapsto \min(\sqrt{2}-1, f_p(x))$ with the cap acting as a reset. The central radius takes only the values $3-2\sqrt{2}$ and $\sqrt{2}-1$ for $6 \leq N \leq 6000$, except $r_9 = r_{17} = 1/3$ (Figure 2). The value $\sqrt{2}-1$ recurs with periods $2, 3, 4, \dots, 7$ that lock in over long ranges of N , last occurs at $N = 597$, and never occurs for $598 \leq N \leq 6000$.
5. *A conjecture and its obstruction* (Section 7). We conjecture that $r_N = 3-2\sqrt{2}$ for every $N \geq 598$. The natural proof strategy, coupling all possible histories from far below the top, fails as stated: the threshold has a square-root edge, so some initial radii are never absorbed. The conjecture amounts to showing the actual orbit avoids these exceptional sets.

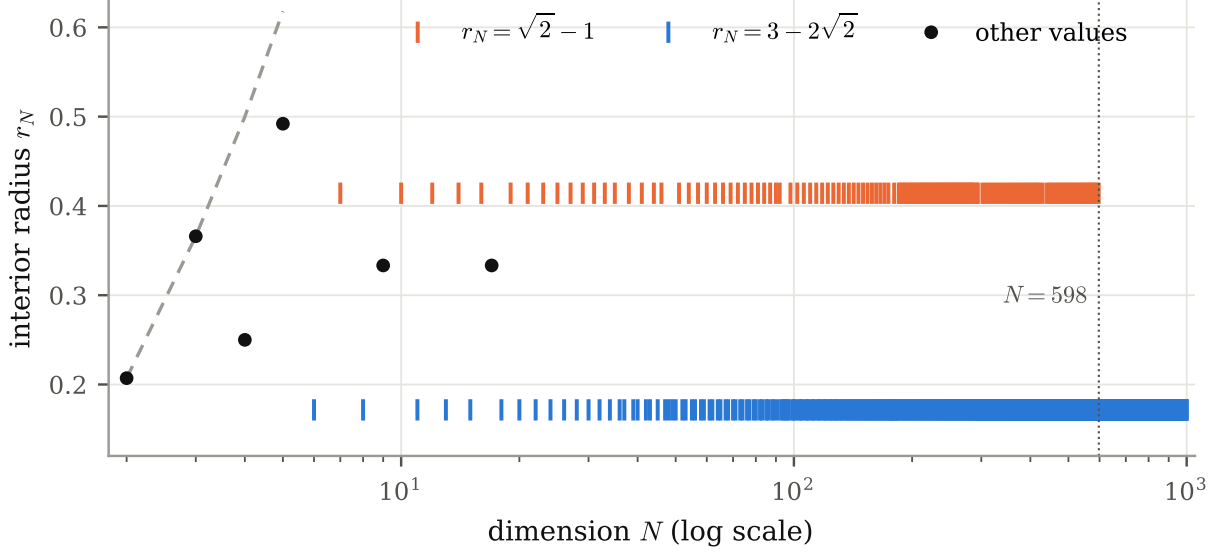


Figure 2: The central radius r_N for $2 \leq N \leq 1000$ (full recursion). Dashed: the corner-only radius $(\sqrt{N} - 1)/2$, which leaves the plot near $N = 5$. For $N \geq 6$ the central radius is $3 - 2\sqrt{2}$ or $\sqrt{2} - 1$, except at $N = 9$ and $N = 17$; to the right of the dotted line ($N \geq 598$) it is always $3 - 2\sqrt{2}$.

2 The construction

Let $C = [-1, 1]^N$ with $N \geq 2$. For $0 \leq k \leq N$, a k -face is determined by a set $F \subseteq \{1, \dots, N\}$ of k free coordinates and a sign vector $\sigma \in \{\pm 1\}^{\{1, \dots, N\} \setminus F}$; it is the set of points of C with $x_i = \sigma_i$ for $i \notin F$.

Definition 1 (Face balls and the growth rule). The ball attached to the k -face (F, σ) with radius r is the N -dimensional ball of radius r centered at the point c with $c_i = 0$ for $i \in F$ and $c_i = \sigma_i(1 - r)$ for $i \notin F$. It is tangent to the $N - k$ facets that contain the face. Faces are processed in order $k = 0, 1, \dots, N$. At stage k all balls on k -faces are grown simultaneously from radius 0, and each stops at its first contact with (i) the boundary of C , (ii) a ball placed at an earlier stage, or (iii) another ball of stage k . Equivalently, the radius at stage k is the right endpoint of the connected component containing 0 of the set of admissible radii.

The phrase “first contact” matters: for some pairs of balls the non-overlap condition holds again for large radii, beyond an interval of overlap. A ball grown from 0 cannot jump that interval, so such radii are excluded.

3 The recursion

Write $a_k = 1 - r_k$ for the offset of a k -face center along each of its fixed coordinates. Throughout, $p = N - k$ and, for $j < k$, $s = k - j$.

Lemma 2 (Symmetry and containment). *The signed permutations of coordinates preserve C and act transitively on k -faces for each k , so all balls of stage k receive the same radius r_k . A k -face ball of radius $r \leq 1$ lies in C .*

Proof. The group of signed coordinate permutations maps faces to faces of the same dimension, transitively, and maps the configuration of all earlier stages to itself; hence every ball of stage k faces the same constraints. For containment, along a free coordinate the center is at distance 1 from both walls; along a fixed coordinate it is at distance r from the near wall and $2 - r$ from the far wall. So the ball lies in C if and only if $r \leq 1$. $\square$

Lemma 3 (Nearest lower-layer ball). *Let $j < k$ and $r_j, r_k < 1$. The minimum distance between the center of a k -face ball and the center of a j -face ball is D_{kj} , where*

$$D_{kj}^2 = (N - k)(r_j - r_k)^2 + (k - j)(1 - r_j)^2.$$

Proof. By symmetry fix the k -face center to have its $N - k$ fixed coordinates equal to a_k and its free coordinates equal to 0. A j -face fixes $N - j$ coordinates. Let x of them be among the k -face's fixed coordinates; the remaining $y = (N - j) - x$ lie among its free coordinates. A shared fixed coordinate contributes $(a_k - a_j)^2$ with equal signs and $(a_k + a_j)^2$ otherwise, so equal signs are optimal and the squared distance is

$$C(x) = x(a_k - a_j)^2 + (N - k - x)a_k^2 + (N - j - x)a_j^2.$$

Since $C'(x) = (a_k - a_j)^2 - a_k^2 - a_j^2 = -2a_k a_j < 0$, C is minimized by the largest admissible x , namely $x = N - k$ (then $y = k - j \leq k$). Substituting, and using $a_k - a_j = r_j - r_k$, gives the formula. $\square$

Lemma 4 (Same-layer cap). *For $1 \leq k \leq N - 1$, the minimum distance between centers of two distinct k -face balls is $\sqrt{2}a_k$, and the same-layer constraint is $r_k \leq \sqrt{2} - 1$. For $k = 0$ the minimum distance is $2a_0$, giving $r_0 = 1/2$.*

Proof. If two k -faces have the same free set, they differ in the sign of at least one fixed coordinate, and their centers are at distance at least $2a_k$. If the free sets $F_1 \neq F_2$ differ, let $m = |F_1 \setminus F_2| = |F_2 \setminus F_1| \geq 1$. Each of the $2m$ coordinates that is free in one face and fixed in the other contributes a_k^2 , and with matching signs on the common fixed coordinates the squared distance is $2m a_k^2$, minimized at $m = 1$. Such a single swap of a free and a fixed coordinate exists when $1 \leq k \leq N - 1$, and $\sqrt{2}a_k < 2a_k$. Tangency $\sqrt{2}(1 - r_k) = 2r_k$ gives $r_k = \sqrt{2}/(2 + \sqrt{2}) = \sqrt{2} - 1$. For $k = 0$ no swap exists and $2a_0 = 2r_0$ gives $r_0 = 1/2$; containment holds since $r_0 \leq 1$. $\square$

Lemma 5 (First-contact bound from one lower layer). *Let $j < k$, $p = N - k$, $s = k - j$, and fix $r_j \in (0, 1)$. The admissible radii r_k for which the k -ball does not overlap the nearest j -ball, and which are reachable by growth from 0, form the interval $[0, \beta(k, j)]$, where*

$$\beta(k, j) = \begin{cases} \frac{(p+1)r_j - \sqrt{4pr_j^2 - (p-1)s(1-r_j)^2}}{p-1}, & p \geq 2, \ 4pr_j^2 \geq (p-1)s(1-r_j)^2, \\ +\infty, & p \geq 2, \ 4pr_j^2 < (p-1)s(1-r_j)^2, \\ \frac{s(1-r_j)^2}{4r_j}, & p = 1, \\ \sqrt{s}(1-r_j) - r_j, & p = 0. \end{cases}$$

Proof. Overlap occurs when $D_{kj} < r_k + r_j$. Set $Q(r_k) = D_{kj}^2 - (r_k + r_j)^2$. By Lemma 3,

$$Q(r_k) = (p-1)r_k^2 - 2(p+1)r_j r_k + (p-1)r_j^2 + s(1-r_j)^2.$$

For $p \geq 2$, Q is an upward parabola with vertex at $r_k = (p+1)r_j/(p-1) > 0$, and $Q(0) = (p-1)r_j^2 + s(1-r_j)^2 > 0$. Its discriminant is $4[4pr_j^2 - (p-1)s(1-r_j)^2]$. If this is negative, $Q > 0$ everywhere and the j -balls never constrain the k -ball. Otherwise both roots are positive, $\{Q \geq 0\} = [0, \rho_1] \cup [\rho_2, \infty)$, and growth from 0 stops at the smaller root ρ_1 , which is the stated formula. For $p = 1$, $Q(r_k) = -4r_j r_k + s(1-r_j)^2$ is decreasing and vanishes at the stated value. For $p = 0$ ($k = N$), $D_{Nj} = \sqrt{N-j}(1-r_j)$ and $Q \geq 0$ means $r_N \leq \sqrt{s}(1-r_j) - r_j$. $\square$

Theorem 6 (The recursion). *For every $N \geq 2$, the radii produced by the growth rule of Definition 1 are $r_0 = \frac{1}{2}$,*

$$r_k = \min\left(\sqrt{2}-1, \min_{0 \leq j < k} \beta(k, j)\right) \quad (1 \leq k \leq N-1), \quad r_N = \min_{0 \leq j \leq N-1} \left(\sqrt{N-j}(1-r_j) - r_j\right),$$

with β as in Lemma 5. Each r_k is determined by $r_0, \dots, r_{k-1}$.

Proof. By Lemma 2 each stage has a single radius. At stage k the admissible set is the intersection of the constraints from the boundary, from each earlier layer and from the same layer. By Lemmas 4 and 5, the connected component containing 0 of each individual constraint set is an interval $[0, b]$, so the component containing 0 of the intersection is $[0, \min b]$. The boundary constraint $r_k \leq 1$ never binds: for $1 \leq k \leq N-1$ the cap gives $r_k \leq \sqrt{2}-1 < 1$, and for $k = N$ the term $j = N-1$ gives $r_N \leq 1 - 2r_{N-1} < 1$. There is no same-layer constraint at $k = N$, since there is a single central ball. Finally, the binding j -ball for a k -ball is the nearest one, since all j -balls have the same radius, and Lemma 3 gives its distance. $\square$

Remark 7. For $N = 2$ and $N = 3$ the central radius equals the corner-only value $(\sqrt{N}-1)/2$: the binding constraint is still the corner layer. From $N = 4$ on, the face layers bind first.

4 Small dimensions

Table 1 lists exact central radii for small N , computed from Theorem 6 in exact arithmetic. The comparison column is the corner-only radius.

The two recurring values have a simple origin. If the top face layer is capped, $r_{N-1} = \sqrt{2}-1$, the term $j = N-1$ gives $r_N \leq 1 - 2(\sqrt{2}-1) = 3 - 2\sqrt{2}$. If instead layer $N-2$ is capped, the term $j = N-2$ gives $r_N \leq \sqrt{2}(2-\sqrt{2}) - (\sqrt{2}-1) = \sqrt{2}-1$. Both are two-dimensional numbers: the binding contacts involve only one or two coordinates, whatever N is.

The same two terms bound the central radius in every dimension.

Proposition 8. *For every $N \geq 2$, $r_N < 1/2$.*

Proof. For $N \leq 4$ this is read off Table 1. Let $N \geq 5$, write $c = \sqrt{2}-1$, and note $c^2 = 3 - 2\sqrt{2}$ and $(1-c)^2 = 2c^2$. Let $x = r_{N-2}$; by Theorem 6, $0 < x \leq c$, and likewise $r_j \leq c$ for $1 \leq j \leq N-1$.

Step 1: the top face layer. Layer $N-1$ has $p = 1$, so by Lemma 5 its bound from layer j is $s(1-r_j)^2/(4r_j)$, which is decreasing in r_j . For $j = N-2$ ($s = 1$) the bound is $q(x) = (1-x)^2/(4x)$. For $1 \leq j \leq N-3$ we have $s \geq 2$ and $r_j \leq c$, so the bound is at least $2(1-c)^2/(4c) = c$. For $j = 0$ the bound is $(N-1)/8 \geq \frac{1}{2} > c$. Hence

$$r_{N-1} = \min(c, q(x)).$$

N	r_N (exact)	numerical	$(\sqrt{N} - 1)/2$
2	$(\sqrt{2} - 1)/2$	0.207107	0.207107
3	$(\sqrt{3} - 1)/2$	0.366025	0.366025
4	$1/4$	0.250000	0.500000
5	$(\sqrt{10} + \sqrt{5} - \sqrt{2} - 3)/2$	0.492066	0.618034
6	$3 - 2\sqrt{2}$	0.171573	0.724745
7	$\sqrt{2} - 1$	0.414214	0.822876
8	$3 - 2\sqrt{2}$	0.171573	0.914214
9	$1/3$	0.333333	1.000000
10	$\sqrt{2} - 1$	0.414214	1.081139
11	$3 - 2\sqrt{2}$	0.171573	1.158312
12	$\sqrt{2} - 1$	0.414214	1.232051
17	$1/3$	0.333333	1.561553

Table 1: Exact central radii for small N .

Step 2: the center. The terms $j = N - 1$ and $j = N - 2$ of Theorem 6 give

$$r_N \leq 1 - 2r_{N-1} = \max(3 - 2\sqrt{2}, g(x)), \quad r_N \leq h(x),$$

where $g(x) = 1 - (1 - x)^2/(2x)$ and $h(x) = \sqrt{2} - (1 + \sqrt{2})x$. Since $3 - 2\sqrt{2} < \frac{1}{2}$, it suffices to show $\min(g(x), h(x)) < \frac{1}{2}$.

Step 3: the two thresholds overlap. We have $g(x) < \frac{1}{2}$ if and only if $(1 - x)^2 > x$, that is, $x < (3 - \sqrt{5})/2 \approx 0.38197$; and $h(x) < \frac{1}{2}$ if and only if $x > (5 - 3\sqrt{2})/2 \approx 0.37868$. The second threshold is below the first: $5 - 3\sqrt{2} < 3 - \sqrt{5}$ is equivalent to $(2 + \sqrt{5})^2 < 18$, that is, $4\sqrt{5} < 9$, that is, $80 < 81$. So every $x \in (0, c]$ satisfies at least one of the two conditions, and $r_N < \frac{1}{2}$. $\square$

Remark 9. The bound in the proof is at most about 0.4956, attained near $x \approx 0.3805$, while the largest central radius in Table 1 is $r_5 \approx 0.4921$. So the constant $1/2$ is close to sharp for this argument. By contrast, the corner-only radius $(\sqrt{N} - 1)/2$ equals $1/2$ at $N = 4$ and exceeds it for all larger N .

5 The step map

The bound that layer $k - 1$ places on layer k (the case $s = 1$ of Lemma 5) defines, for $p \geq 2$,

$$f_p(x) = \frac{(p+1)x - \sqrt{d_p(x)}}{p-1}, \quad d_p(x) = 4px^2 - (p-1)(1-x)^2,$$

on the set where $d_p(x) \geq 0$, that is, $x \geq x^*(p)$ with

$$x^*(p) = \frac{\sqrt{p-1}}{2\sqrt{p} + \sqrt{p-1}}.$$

For $x < x^*(p)$, layer $k - 1$ does not constrain layer k at all.

Proposition 10. *Let $p \geq 2$.*

(a) $f_p(1/3) = 1/3$ and $f'_p(1/3) = -2$.

(b) For $x \geq x^*(p)$, $f_p(x) < x$ if and only if $x > 1/3$.

(c) $x^*(p) < 1/3$, $x^*(p)$ increases to $1/3$, and $1/3 - x^*(p) \sim 1/(9p)$ as $p \rightarrow \infty$.

Proof. (a) $d_p(1/3) = \frac{4p}{9} - \frac{4(p-1)}{9} = \frac{4}{9}$, so $f_p(1/3) = (\frac{p+1}{3} - \frac{2}{3})/(p-1) = \frac{1}{3}$. Since $d'_p(x) = 8px + 2(p-1)(1-x)$, we get $d'_p(1/3) = (12p-4)/3$ and $f'_p(1/3) = ((p+1) - d'_p(1/3)/(2\sqrt{d_p(1/3)}))/(p-1) = ((p+1) - (3p-1))/(p-1) = -2$.

(b) Since $p-1 > 0$, $f_p(x) < x$ is equivalent to $2x < \sqrt{d_p(x)}$, and both sides are nonnegative, so to $4x^2 < 4px^2 - (p-1)(1-x)^2$, that is, to $(p-1)(4x^2 - (1-x)^2) > 0$. Since $4x^2 - (1-x)^2 = (3x-1)(x+1)$, this holds if and only if $x > 1/3$.

(c) Write $q = \sqrt{(p-1)/p} \in (0, 1)$, so $x^*(p) = q/(2+q)$, which is increasing in q and equals $1/3$ at $q = 1$. Then $1/3 - x^*(p) = 2(1-q)/(3(2+q))$, and $1-q \sim 1/(2p)$. $\square$

So a run of layers each pinned by the one below descends monotonically while it stays above $1/3$ (part (b)); near $1/3$ the deviation is doubled and reflected (part (a)); and a radius that lands below $x^*(p)$, within about $1/(9p)$ under $1/3$, releases the next layer, which then grows to the cap. Figure 3(b) shows this cycle. The cap is an exact reset: two different histories that reach the cap on the same layer agree on every layer above it.

Corollary 11. For $N = 17$, $r_k = 1/3$ for $1 \leq k \leq 16$ and $r_{17} = 1/3$.

Proof. With $p = 16$, $s = 1$ and $r_0 = \frac{1}{2}$, Lemma 5 gives $d = 16 - \frac{15}{4} = \frac{49}{4}$ and $\beta(1, 0) = (\frac{17}{2} - \frac{7}{2})/15 = \frac{1}{3}$, which is below the cap, so $r_1 = 1/3$. (For $p \geq 2$, the equation $f_p(1/2) = 1/3$ reduces to $p^2 - 17p + 16 = 0$, whose only root with $p \geq 2$ is $p = 16$; so $N = 17$ is the only dimension in which the first face layer lands exactly on the fixed point.)

Now let $2 \leq k \leq 16$, so $p = 17 - k \geq 1$, and suppose $r_1 = \dots = r_{k-1} = \frac{1}{3}$. We check every bound on r_k from Theorem 6.

- $j = k - 1$ ($s = 1$). For $p \geq 2$ the bound is $f_p(\frac{1}{3}) = \frac{1}{3}$ by Proposition 10(a); for $p = 1$ it is $(\frac{2}{3})^2/(4 \cdot \frac{1}{3}) = \frac{1}{3}$.
- $1 \leq j \leq k - 2$ ($s \geq 2$, $r_j = \frac{1}{3}$). For $p \geq 2$ the discriminant condition of Lemma 5 reads $\frac{4}{9}(p - (p-1)s) \geq 0$, but $p - (p-1)s \leq 2 - p \leq 0$. So either there is no constraint, or ($p = s = 2$) the double root $(p+1)r_j/(p-1) = 1$ is the bound. For $p = 1$ the bound is $s/3 \geq \frac{2}{3}$.
- $j = 0$ ($s = k \geq 2$, $r_0 = \frac{1}{2}$). For $p \geq 2$ the discriminant condition reads $\frac{1}{4}(p - (p-1)k) \geq 0$, and $p - (p-1)k \leq 2 - p \leq 0$, with equality only when $p = k = 2$, that is, $N = 4$. So there is no constraint. For $p = 1$ ($k = 16$) the bound is $16/8 = 2$.
- The cap $\sqrt{2} - 1$ exceeds $\frac{1}{3}$.

Hence $r_k = \frac{1}{3}$, and by induction $r_1 = \dots = r_{16} = \frac{1}{3}$. For the center, the term $j = 16$ gives $1 - 2 \cdot \frac{1}{3} = \frac{1}{3}$, while the terms $1 \leq j \leq 15$ give $\frac{2}{3}\sqrt{17-j} - \frac{1}{3} \geq \frac{2}{3}\sqrt{2} - \frac{1}{3} > \frac{1}{3}$ and $j = 0$ gives $(\sqrt{17} - 1)/2 > \frac{1}{3}$. So $r_{17} = \frac{1}{3}$. $\square$

For $N = 9$ the orbit also reaches $1/3$ exactly, at layer 2, and stays there.

6 Numerical results

All computations below use Theorem 6 directly, checking every lower layer $j < k$, in double precision. The cases $N \in \{9, 17, 581, 588, 597, 598, 600\}$ were recomputed in 60-digit arithmetic with identical results; there, the winning bound for r_N beats the runner-up by at least 0.17, and every near-zero margin between layer bounds is an exact tie between two constraints giving the same value. Code is available at the repository listed at the end.

Observation 12 (Nearest-layer reduction). For every $16 \leq N \leq 6000$ and every $1 \leq k \leq N - 1$, r_k is either the cap $\sqrt{2} - 1$ or $\beta(k, k - 1)$. Equivalently, the recursion reduces to

$$r_k = g_{N-k}(r_{k-1}), \quad g_p(x) = \begin{cases} \sqrt{2} - 1, & x < x^*(p), \\ \min(\sqrt{2} - 1, f_p(x)), & x \geq x^*(p), \end{cases}$$

for $p \geq 2$, with $g_1(x) = \min(\sqrt{2} - 1, (1 - x)^2/(4x))$. For $10 \leq N \leq 15$ the reduction fails at layer $k = 2$, where the corner layer ($s = 2$) binds, but the central radius agrees with the reduced map.

In the reduced picture, $r_N = \min(1 - 2r_{N-1}, \sqrt{2}(1 - r_{N-2}) - r_{N-2})$ in the observed range, and the central radius is $3 - 2\sqrt{2}$ exactly when the top face layer is capped.

Observation 13 (Two values and mode locking). For $6 \leq N \leq 6000$, $r_N \in \{3 - 2\sqrt{2}, \sqrt{2} - 1\}$ except $r_9 = r_{17} = 1/3$. The value $\sqrt{2} - 1$ occurs at

- $N = 19, 21, \dots, 35$ (every second N),
- $N = 51, 54, \dots, 90$ (every third),
- $N = 98, 102, \dots, 174$ (every fourth),
- $185 \leq N \leq 287$, with gaps alternating 2, 3 (two in every five),
- $N = 302, 308, \dots, 422$ (every sixth),
- $N = 441, 448, \dots, 588$ (every seventh),

with short transitional stretches between these ranges (for example $N = 7, 10, 12, 14, 16$ and $N = 38, 41, 44, 46$). The last occurrence is $N = 597$. For $598 \leq N \leq 6000$, $r_N = 3 - 2\sqrt{2}$.

Observation 14 (Exact merging at the cap). Read from the top, the layer radii converge to a single profile $R(p)$ that does not depend on N . Comparing with $N = 300,000$ (computed with the reduced map), the profiles agree exactly on the top 282 layers for $N = 598$, the top 988 for $N = 1000$, and the top 29,319 for $N = 50,000$. In each case the deepest agreeing layer is a cap shared by both, and just below it the two profiles differ by less than 10^{-3} (by about 6×10^{-6} for $N = 50,000$). For $N = 597$ the profile has not merged when it reaches the top.

Figure 4 shows the same phenomenon across all $N \leq 800$: near the top of the stack the pattern of capped layers is aligned with $p = N - k$, and from $N = 598$ on the top two dozen layers are the same for every N .

Beyond the range of the full recursion we use the reduced map, which is linear-time per dimension. For 60 randomly chosen N between 6001 and 300,000 it gives a central radius of $3 - 2\sqrt{2}$. These runs assume that Observation 12 continues to hold beyond $N = 6000$.

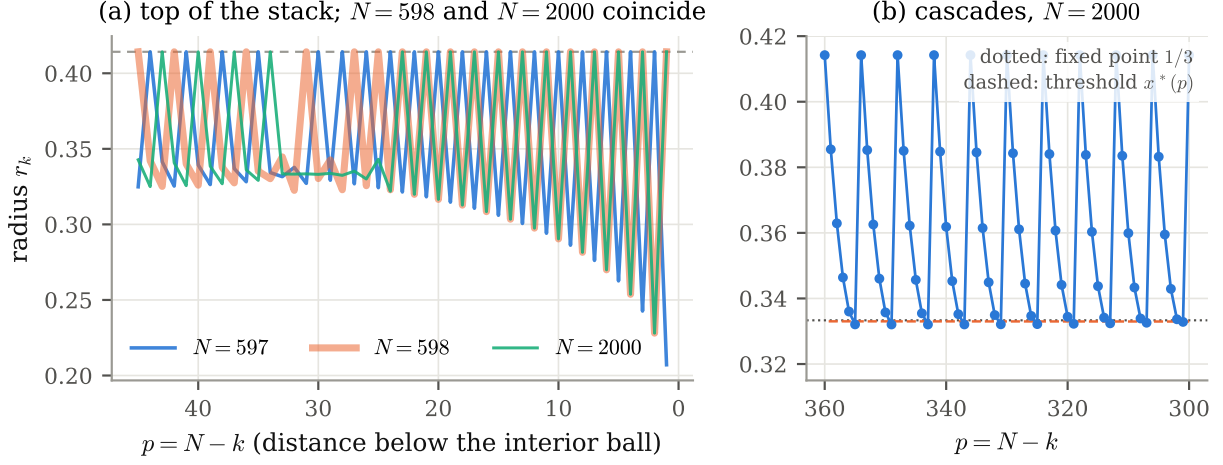


Figure 3: (a) The top 45 layers for $N = 597, 598$ and 2000 , plotted against $p = N - k$. Layers alternate between the cap and a radius pinned by the layer below. For $N = 597$ the top face layer ($p = 1$) is pinned, giving $r_N = \sqrt{2} - 1$; for $N = 598$ and $N = 2000$ it is capped, giving $3 - 2\sqrt{2}$, and these two curves are identical over the whole window. (b) Cascades deep in the stack for $N = 2000$: each curve descends towards the fixed point $1/3$ and resets to the cap once a radius falls below the threshold $x^*(p)$.

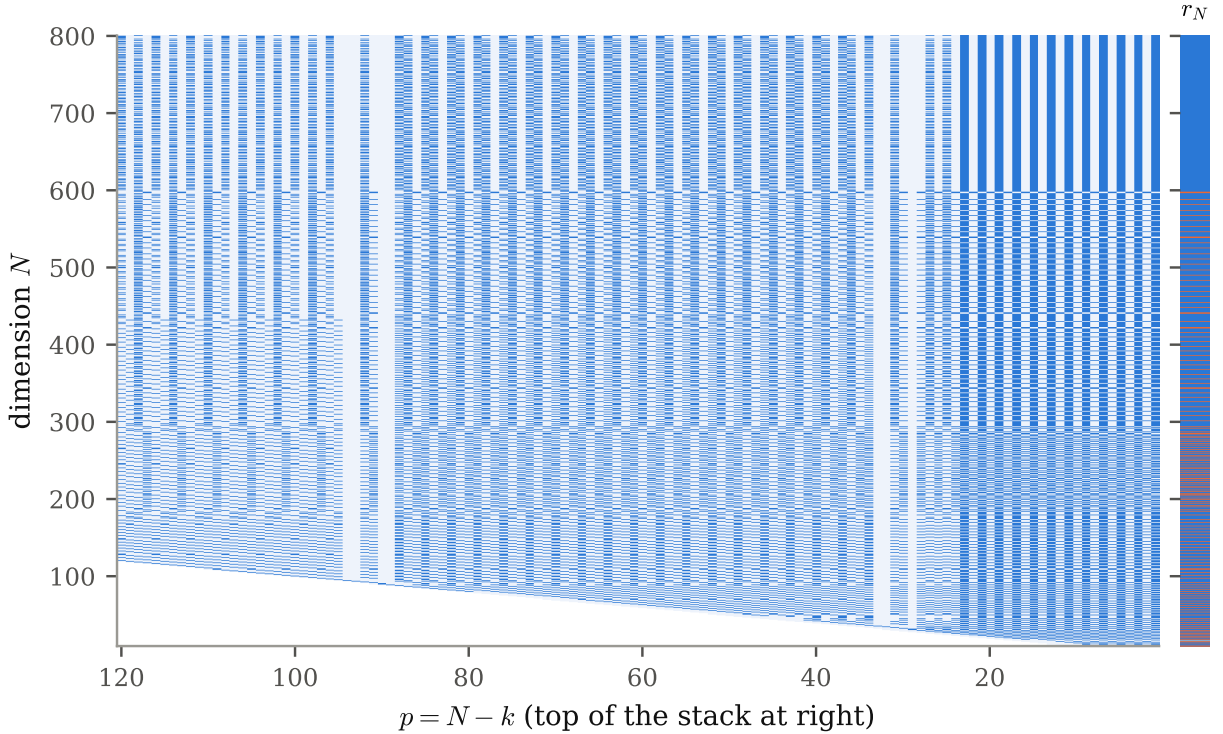


Figure 4: Capped layers (dark) for $10 \leq N \leq 800$, aligned at the top of the stack ($p = N - k$, top at right). The strip at right marks the central radius: $3 - 2\sqrt{2}$ (blue) or $\sqrt{2} - 1$ (orange). From $N = 598$ the top columns become identical in every row and the strip is uniformly blue.

7 Conjectures and the obstruction to a simple proof

Conjecture 15. *For every $N \geq 16$ and $1 \leq k \leq N - 1$, $r_k \in \{\sqrt{2} - 1, \beta(k, k - 1)\}$.*

Conjecture 16. *For every $N \geq 598$, $r_N = 3 - 2\sqrt{2}$.*

By Proposition 8, the central radius is already known to lie in $(0, \frac{1}{2})$ for every N ; the conjecture asserts that it eventually takes a single value.

A natural approach to Conjecture 16 is coupling from the past [2]: since the map g_p depends only on p , the state at depth $p = D$ below the top lies in some set S_D for every $N > D$, and if the image of S_D under $g_{D-1}, \dots, g_1$ collapsed to a single orbit, every $N > D$ would be settled at once. This fails as stated. Pushing the whole interval $(0, \frac{1}{2}]$ down from depths $D = 600, 1000, 3000$ and 20,000, the set of reachable radii at the top face layer is the entire interval $[\frac{\sqrt{2}-1}{2}, \sqrt{2} - 1]$. Images of intervals under the continuous pieces of g_p are computed exactly here, so this is not an artifact of over-approximation. The cause is the square-root edge of f_p at $x^*(p)$, where f_p has infinite slope: a thin window of radii just above the threshold is stretched over a whole range.

The exceptional set is nevertheless small. Of 20,000 evenly spaced starting radii in $(0, \frac{1}{2}]$ at depth $D = 600$, 97.8% end with $r_N = 3 - 2\sqrt{2}$; at depth $D = 2000$ all of them do. So Conjecture 16 is a statement about the particular orbit that starts from $r_0 = \frac{1}{2}$: it must avoid the exceptional windows near the thresholds. For $598 \leq N \leq 3000$ the closest approach of an actual layer radius to its threshold is about 3×10^{-8} , at $N = 2464$, $k = 14$, deep in the stack where later merging can absorb it.

Questions.

1. Prove Conjecture 15, or find the first N where another layer binds.
2. Describe the exceptional sets: their location and width as functions of the depth D , and whether their measure tends to 0.
3. Explain the mode-locked periods of Observation 13. Proposition 10 suggests that a cascade starting from the cap needs roughly $\sqrt{p(f_p(\sqrt{2} - 1) - \frac{1}{3})}$ steps to reach $1/3$, since steps far from $1/3$ have size about $2\sqrt{(x - \frac{1}{3})/p}$; how does this produce the observed sequence of periods?
4. Is there an analogous threshold-and-reset reduction for other greedy constructions with large symmetry groups, such as greedy packings in the cross-polytope or simplex?

Code availability. Python code reproducing every computation and figure is available at <https://github.com/DanielKetterer/hypercube-face-balls>.

References

- [1] R. W. Hamming, *The Art of Doing Science and Engineering: Learning to Learn*, Gordon and Breach, Amsterdam, 1997, Chapter 9.
- [2] J. G. Propp and D. B. Wilson, Exact sampling with coupled Markov chains and applications to statistical mechanics, *Random Structures Algorithms* **9** (1996), 223–252.